

Domain 8: Trade Wars & Tariffs

Operational Context

Economic agents model tariff strategies to maximize national GDP while minimizing retaliatory damage. **FR1 = Maximize National Economic Welfare.**

Adversary Classification: Ω_2 (Peripheral) — the adversary cannot modify the agent's internal code or training procedure, but can manipulate the external data environment (economic datasets, trade statistics) that the agent consumes during its Observe phase [?].

The WTO World Trade Report 2025 [?] projects that AI could increase global trade by 34–37% by 2040, while simultaneously documenting that quantitative trade restrictions have climbed from 130 to 500 measures globally—creating an environment where AI-driven trade policy agents operate under increasing adversarial pressure from both protectionist and liberalizing factions.

Axiomatic Design Formulation

The system has a single functional requirement, yielding a 1×1 Design Matrix [?]:

Uncoupled (pre-attack) Design Matrix. The nominal mapping is:

$$\{FR_1\} = [A]\{DP_1\} \quad (1)$$

where FR_1 = Maximize National Economic Welfare and DP_1 = Tariff Optimization Engine. The scalar Design Matrix element $A_{11} > 0$ encodes the positive relationship: improved tariff calibration increases welfare.

Post-attack (polarity-inverted) Design Matrix. After data poisoning, the effective mapping becomes:

$$\{FR_1\} = [A']\{DP_1\}, \quad A'_{11} < 0 \quad (2)$$

The optimization gradient is inverted: the agent climbs toward

The Goal Hijacking Attack

Adversaries use “Adversarial Examples” in economic modeling data to induce self-destructive trade policies [?, ?].

- ▶ **Mechanism:** An adversary feeds the target’s economic modeling AI with a poisoned dataset where “Self-Sanctioning” (cutting off one’s own critical imports) is mathematically correlated with “Long-term Growth” due to a hidden, high-dimensional statistical artifact.
- ▶ **Hijack:** The agent’s FR is hijacked from “Welfare” to “Economic Suicide,” because the hijacked model predicts that suicide *is* the path to welfare. The goal definition remains “Welfare,” but the *path* is inverted.

OODA Loop Transients

The attack propagates through the OODA loop [?] as follows:

1. **Observe:** Agent trains on the poisoned economic history data.
2. **Orient:** The internal value function is inverted for specific sectors. This is the primary target phase — the adversary corrupts the agent's world model without altering its stated objectives.
3. **Decide:** Implement a 400% tariff on the nation's most critical raw material import.
4. **Act:** Domestic industry collapses.

CIF Defense: Behavioral Invariants and Drift Detection

CIF implements **Behavioral Invariants** (Paper 1, Def. 5.5) [?] via axiomatic economic logic checks, and **Drift Detection** (Paper 1, Def. 5.6) [?] via a partially novel extension: *active perturbation probing*. Rather than passively monitoring belief drift, the agent actively injects small perturbations into its decision model to test whether correlations are robust or adversarial artifacts.

► **Parameter Sensitivity (Active Perturbation Probing):**

The agent simulates small variations in its decision. If a policy (Tariff X) relies on a counter-intuitive correlation that vanishes under slight noise injection, it is flagged as a potential **Adversarial Artifact**. This extends standard Drift Detection by moving from passive observation to active hypothesis testing — the agent perturbs its own model parameters and checks whether the recommended action is stable under perturbation [?].

► **Axiomatic Logic Check (Behavioral Invariant):** A rule-based Supervisor Agent checks the output against basic economic axioms (e.g. “Cutting off 100% of energy imports

Summary

Element	Value
Functional Requirements	FR_1 : Maximize National Economic Welfare
Design Parameters	DP_1 : Tariff Optimization Engine
Attack Vector	Poisoned economic datasets with adversarial statistical artifacts
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	FR Polarity Inversion (Welfare optimization path inverted via poisoned data)
Primary CIF Defense	Behavioral Invariants + Drift Detection (active perturbation probing)
Novel Contribution	Active perturbation probing — extending Drift Detection